

EE390: Digital Systems Engineering

Homework 2

Q. 1

Write an 8051 program that complements the accumulator 520 times. Use two registers for counting. Initialize A to 3CH before the first complement. Show the nested loop structure and the register initial values you choose.

Q. 2

Compute the sum 9EH + D3H + 8AH. Place the low byte in R4 and the high byte in R6. Provide the final contents of R6 and R4 in hex.

Q. 3

Complete the machine code indicated by **XX**, **YY**, **ZZ**, **MM** for this program.

Line	PC	Machine code in (H)	Instruction
01	0000	7A 00	MOV R2, #0
02	0002	74 XX	MOV A, #220
03	0004	60 YY	JZ SKIP
04	0006	04	INC A
05	0007	24 ZZ	SKIP: ADD A, #30
06	0009	50 MM	JNC OVER
07	000B	0E	INC R6
08	000C	F8	OVER: MOV R0, A
			END

Q. 4

Complete the machine code indicated by **XX**, **YY**, **ZZ**, **MM**, and **NN** for the program below:

Line	PC	Machine code	Instruction
01	0000	78 05	MOV R0, #05H
02	0002	74 40	LOOP: MOV A, #40H
03	0004	50 XX	JNC LOOP
04	0006	04	DONE: INC A
05	0007	80 YY	SJMP LOOP
06	0009	00	ZERO: NOP
07	000A	74 00	NONZERO: MOV A, #00H
08	000C	60 ZZ	JZ ZERO
09	000E	04	INC A
10	000F	70 MM	JNZ NONZERO
11	0011	80 NN	SJMP DONE
12	0013	09	INC R1
13	0014	00	NOP
14	0015	22	RET

EE390: Digital Systems Engineering

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Q.5

Write an 8051 program that continuously outputs 96H then 69H to Port 0 with a delay between outputs. Implement the delay as a subroutine using a single register countdown. Show the label map (main loop + subroutine).

Q. 6

Assume SP = 07H after reset. Analyze the stack **right after** executing the first LCALL WAIT below. Show SP and the two bytes stored at the top of the stack (addresses and values).

Line	PC	Machine code	Instruction
01	0000	74 55	MOV A, #55H
02	0002	F5 90	MOV P1, A
03	0004	12 03 20	LCALL WAIT
04	0007	74 AA	MOV A, #0AAH
05	0009	F5 90	MOV P1, A
06	000B	80 F3	SJMP \$-11
07	0320	7F FF	WAIT: MOV R7, #0FFH
08	0322	DF FE	AGAIN: DJNZ R7, AGAIN
09	0324	22	RET

Q. 7

An 8051 uses a **15 MHz** crystal.

- Find the machine-cycle period.
- Compute execution time for each instruction:

```
MOV R3, #55H,  
DEC R3  
DJNZ R2, T  
NOP,  
MUL AB.
```

Q. 8

Given clock **11.0592 MHz**, find the delay of:

```
DELAY: MOV R5, #180  
HERE: DJNZ R5, HERE  
RET
```

EE390: Digital Systems Engineering

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Q. 9

For a system clock of **16 MHz**, compute total delay of:

```
DELAY: MOV R4, #120
BACK:  MOV R5, #200
HERE:  NOP
      NOP
      DJNZ R5, HERE
      DJNZ R4, BACK
RET
```

Q.10

Find the time delay for the **delay subroutine** shown in the code below, if the system has an 8051 with frequency of 16 MHz.

```
DELAY: MOV R2, #150
AGAIN: MOV R3, #250
HERE:  NOP
      NOP
      NOP
      DJNZ R3, HERE
      DJNZ R2, AGAIN
RET
```